
Robust Offline Reinforcement Learning via Disentangled State Representations

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1 Introduction

Offline reinforcement learning (offline RL) aims to learn policies from fixed datasets without further interaction with the environment. This setting is important when online exploration is expensive, unsafe, or impractical, but it also makes learning highly sensitive to observation quality. In many realistic scenarios, observations may contain sensor noise, irrelevant nuisance variables, or features that are correlated with the environment but not useful for control. This creates a gap between training-time information and deployment-time observations.

Most standard offline RL methods do not explicitly address this gap, especially when nuisance information is nonlinearly entangled with the true task-relevant state. Directly training a policy on corrupted observations may cause the agent to rely on noisy or irrelevant factors, leading to unstable or suboptimal behavior. This project studies whether disentangled representation learning can improve robustness under such observation corruption.

We propose a Privileged Pretraining Framework (PPF) that pretrains an encoder with clean-state supervision, then freezes it for downstream offline RL on corrupted observations only. This is realistic in control settings where cleaner states can be obtained during dataset construction (e.g., from simulation or post-processed estimation), while deployment must rely on noisier onboard sensors. PPF uses clean states only as training-time privileged information, so the final policy still operates solely on corrupted observations at test time.

We evaluate this framework on D4RL locomotion tasks under three corruption regimes: concat, project, and nonlinear corruption. These settings range from directly appended nuisance dimensions to nonlinear mixing between task-relevant and nuisance information. Within the same PPF backbone, we compare six disentanglement regularizers, analyze when disentanglement improves robustness, and test whether the learned robustness transfers across environments and offline RL backbones. Our work makes the following contributions:

- We formulate a controlled evaluation setting for robust offline RL under observation corruption, including concat, project, and nonlinear corruption regimes.
- We propose a Privileged Pretraining Framework that uses clean states only during encoder pretraining and deploys policies using only corrupted observations.
- We empirically compare six disentanglement regularizers across multiple D4RL environments and show that disentanglement is most beneficial under hard nonlinear corruption.
- We provide cross-environment analysis on HalfCheetah, Hopper, and Ant, showing that disentanglement transfers well to environments with similar state dimensionality, but requires environment-specific tuning of the regularization weight in higher-dimensional environments such as Ant.

2 Related Work

Offline Reinforcement Learning and Robust Offline RL under Corruption. Offline RL is commonly studied with D4RL benchmarks on MuJoCo locomotion tasks [3], using algorithms such as CQL [10], TD3+BC [4], and Implicit Q-Learning (IQL) [9]. We use IQL as the main downstream algorithm because it is strong and stable. However, these methods are not specifically

designed for corrupted-observation settings. Recent work studies robustness in offline RL under data corruption: simple PCA preprocessing serves as a linear noise-removal baseline but does not learn task-aware representations; RIQL [17] introduces robust-statistics-inspired modifications to IQL across diverse corruption types; and Denoised MDP [15] learns task-relevant and task-irrelevant latent components using self-supervised dynamics and reward objectives, but without privileged clean-state supervision. In contrast, our work focuses on *representation-level robustness* with privileged clean-state supervision: learning latent features that separate task-relevant information from nuisance factors under controlled observation corruption.

Representation Learning and Disentanglement. Representation learning aims to learn compact and useful features from high-dimensional observations. Classical disentangled representation learning methods, such as β -VAE [7] and FactorVAE [8], encourage latent variables to capture distinct factors of variation. Self-supervised methods such as Barlow Twins [18], VICReg [1], and whitening-based representation learning [2] reduce redundancy or decorrelate latent dimensions. Dependence-based criteria such as HSIC [5] and distance correlation [13], as well as contrastive objectives such as InfoNCE [11], motivate the regularizers evaluated in our work. However, prior work has not systematically compared these regularizers for robust offline RL under the same controlled corruption setting.

Privileged Information and Partial Observability. Our setting is also related to learning with privileged information, where additional information is available during training but not at test time [14]. In reinforcement learning, asymmetric actor-critic methods use a similar idea by training the critic with full state information while the actor receives only partial observations [12]. Prior work also studies offline RL in partially observable environments, such as ORDER [6] and CO-CQL [16]. Different from standard POMDP settings, we consider a controlled corruption scenario where observations are generated by mixing clean states with nuisance noise. Our main difference is that we combine privileged clean-state supervision with disentangled representation learning to analyze when representation-level disentanglement is useful, unnecessary, or potentially harmful for robust offline RL.

3 Background and Preliminaries

3.1 Offline RL under Observation Corruption

We consider a standard Markov decision process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$, with the goal of learning a policy $\pi(a | s)$ that maximizes the expected discounted return. In offline RL, the agent cannot interact with the environment and instead learns from a fixed dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$, which makes it sensitive to data quality, distribution shift, and observation corruption. While the standard setting assumes clean states, here the agent observes a corrupted observation $\tilde{o}$ instead of the clean state s .

We model corruption as $\tilde{o} = g(s, n; \alpha)$, where n is nuisance noise and α controls the noise scale. Clean states are available only during encoder pretraining as privileged supervision; downstream offline RL training and inference use only corrupted observations.

3.2 Corruption Regimes

To systematically evaluate robustness, we design three corruption regimes with increasing difficulty.

- **Concat:** $\tilde{o} = [s; \alpha n]$. Noise is directly appended to the clean state, so task-relevant and nuisance dimensions remain explicitly separated.
- **Project:** $\tilde{o} = Q[s; \alpha n]$, where $Q^\top Q = I$. A random orthogonal projection linearly mixes task-relevant and nuisance information.
- **Nonlinear:** $\tilde{o} = \tanh(W_2 \tanh(W_1[s; \alpha n] + b_1) + b_2)$. A nonlinear transformation entangles task-relevant and nuisance information, making this the hardest setting.

4 Method

4.1 Privileged Pretraining Framework

Figure 1 illustrates the overall pipeline. Given a corrupted observation $\tilde{o}$, a single encoder maps it to a latent representation $h = f_\theta(\tilde{o}) \in \mathbb{R}^d$.

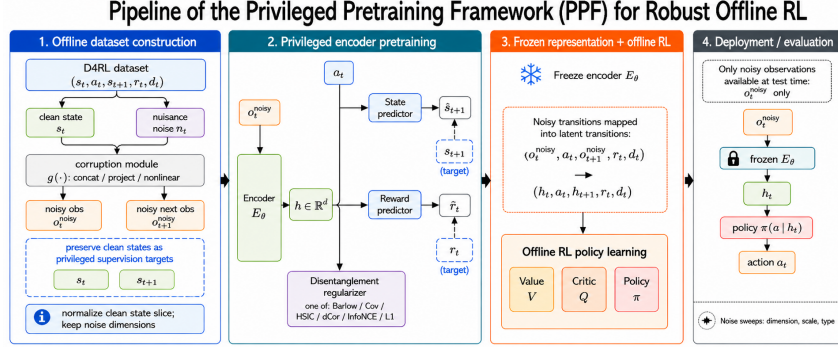


Figure 1: Pipeline of the Privileged Pretraining Framework (PPF) for robust offline RL.

The framework has two stages. First, f_θ is pretrained with privileged clean-state supervision and a disentanglement regularizer on h . Second, f_θ is frozen and used as a fixed feature extractor for downstream offline RL, so clean states act purely as training-time privileged information that is never required at deployment. Intuitively, the disentanglement regularizer prevents nuisance variation from being redundantly distributed across correlated latent dimensions, so that the downstream policy is exposed to a more compact, task-aligned representation.

4.2 Encoder Pretraining Objective

The encoder is trained with three objectives:

$$\mathcal{L}_{\text{pretrain}} = \mathcal{L}_{\text{state}} + \lambda_r \mathcal{L}_{\text{reward}} + \lambda_d \mathcal{L}_{\text{disentangle}}.$$

The state predictor takes h_t and the action a_t and predicts the next clean state, giving a dynamics-aware loss $\mathcal{L}_{\text{state}} = \|\hat{s}_{t+1} - s_{t+1}\|_2^2$. The reward predictor uses h_t to predict the reward, $\mathcal{L}_{\text{reward}} = (\hat{r}_t - r_t)^2$. The disentanglement loss $\mathcal{L}_{\text{disentangle}}$ measures pairwise dependence among the dimensions of h , reducing redundancy or statistical coupling within the latent representation. The weights λ_r and λ_d control the strength of reward prediction and disentanglement regularization, respectively.

4.3 Disentanglement Regularizers

We evaluate six regularizers within the same PPF backbone, all applied to the latent $h \in \mathbb{R}^d$, where C denotes its cross-correlation matrix. These objectives span correlation-based, dependence-based, and contrastive regularization.

- **Barlow Twins:** $\mathcal{L}_{\text{BT}} = \sum_i (1 - C_{ii})^2 + \lambda \sum_{i \neq j} C_{ij}^2$ (reduces redundancy by pushing C toward the identity).
- **Covariance Whitening:** $\mathcal{L}_{\text{cov}} = \sum_{i \neq j} \text{Cov}(h_i, h_j)^2$ (removes linear dependence).
- **HSIC:** $\mathcal{L}_{\text{HSIC}} = \text{HSIC}(h)$ (captures nonlinear statistical dependence).
- **Distance Correlation:** $\mathcal{L}_{\text{dCor}} = \text{dCor}(h)$ (detects general dependence via pairwise distances).
- **InfoNCE:** $\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[\log \frac{\exp(\text{sim}(h, h')/\tau)}{\sum_{h'} \exp(\text{sim}(h, h')/\tau)} \right]$ (contrastive representation learning).
- **L1 Cross-Correlation:** $\mathcal{L}_{\text{L1}} = \sum_{i \neq j} |C_{ij}|$ (encourages sparse coupling).

4.4 Downstream Offline RL Training

After pretraining, the frozen encoder maps the offline dataset into a latent transition dataset $\mathcal{D}_h = \{(h_i, a_i, r_i, h'_i)\}_{i=1}^N$, on which we train Implicit Q-Learning (IQL).

5 Experiments

5.1 Experimental Design

We evaluate on D4RL MuJoCo locomotion datasets: HalfCheetah-medium-v2, Hopper-medium-v2, and Ant-medium-v2. All experiments use synthetic observation corruption, and IQL is used as the main downstream offline RL backbone unless otherwise specified. To ensure fair comparison, we fix the encoder architecture, pretraining budget, downstream training budget, batch size, and random seeds across methods. We compare four method groups: (i) **True Only**, an upper bound (IQL on clean states); (ii) **Raw Noisy**, a lower bound (IQL on corrupted observations); (iii) **Plain Encoder**, the PPF encoder without disentanglement regularization ($\lambda_d = 0$); and (iv) **PPF + (regularizer)**, our PPF with one of the six disentanglement regularizers in Section 4.3.

Main Setup We use HalfCheetah to compare all six regularizers, then evaluate the strongest HalfCheetah methods on Hopper and Ant for cross-environment transfer.

Table 1: Main experimental settings across environments.

Environment	Corruption	Noise Dim.	Noise Scale	Seeds
HalfCheetah Stage 1	Nonlinear	{5, 10, 20, 40, 80}	{0.5, 1, 2, 4, 8}	{1, 2, 3, 5}
HalfCheetah Stage 2	Nonlinear	{4, 8, 13, 17}	{0.5, 1, 1.5, 2}	{1, 2, 3}
HalfCheetah Stage 3	Concat, Project	{17, 34}	{1, 2}	{1, 2, 3}
Hopper	Nonlinear	{3, 6, 8, 11}	{0.5, 1, 1.5, 2}	{1, 2, 3}
Ant	Nonlinear	{28, 56, 83, 111}	{0.5, 1, 1.5, 2}	{1, 2, 3}

Stage 1 is a broad nonlinear stress test that identifies where methods begin to fail. Stage 2 uses a proportional design, $d_{\text{noise}} \in \{0.25d_s, 0.5d_s, 0.75d_s, 1.0d_s\}$, where d_s is the clean-state dimension, so corruption levels are comparable across environments. These controlled nonlinear experiments use noise scales $\{0.5, 1.0, 1.5, 2.0\}$ to cover mild to strong corruption without collapsing all methods. Stage 3 then tests the simpler concat and project corruption types using noise dimensions equal to $1.0\times$ and $2.0\times$ the original HalfCheetah dimension. Based on the HalfCheetah results, we further evaluate the top three methods, HSIC, Barlow Twins, and distance correlation, on Hopper and Ant.

Ablation Setup We use three ablations to identify where the improvement of PPF comes from. All ablations are conducted on HalfCheetah-medium-v2 under nonlinear corruption with noise dimensions $\{4, 8, 13, 17\}$, noise scales $\{0.5, 1.0, 1.5, 2.0\}$, and seeds $\{1, 2, 3\}$. Each ablation keeps the PPF pipeline and changes only one component.

- **No Privileged Supervision.** Replace clean-state supervision with corrupted-observation reconstruction to test whether privileged clean-state targets are necessary.
- **Reward-Only Pretraining.** Remove state and dynamics supervision and keep only reward prediction to test whether reward supervision alone is sufficient.
- **Downstream Backbone.** Replace IQL with TD3+BC while keeping the pretrained encoder fixed. This tests whether the learned representation and disentanglement effect depend specifically on IQL.

External Baselines We compare PPF with no-PPF baselines under the same HalfCheetah-medium-v2 nonlinear-corruption setting, testing whether alternative strategies can replace PPF-style representation learning.

- **PCA-IQL:** applies PCA to noisy observations before IQL training, testing whether simple linear dimensionality reduction is enough.
- **TD3+BC:** trains TD3+BC directly on corrupted observations, testing whether changing the offline RL backbone alone is sufficient.
- **RIQL:** trains a robust IQL variant directly on corrupted observations, testing algorithm-level robustness without PPF-based representation learning.
- **Denoised MDP:** learns task-relevant and task-irrelevant representations with self-supervised objectives, but without privileged clean-state supervision.

Implementation Details. Unless otherwise specified, all methods share the same training budget and IQL settings, as summarized in Table 2.

(a) Disentanglement weights			(b) Common hyperparameters			
Method	indep_weight	Extra Params	Parameter	Value	Parameter	Value
Barlow Twins	0.05	—	Pretrain epochs	50	IQL epochs	100
Cross-Cov.	50.0	—	Pretrain batch	512	IQL batch	256
dCor	3.0	—	Enc. opt.	Adam, 3×10^{-4}	Grad. clip	1.0
HSIC	500.0	$\sigma = 1.0$	Latent dim.	$1.5 \times \max(d_s, d_n)$	Discount	0.99
InfoNCE	1.0	$\tau = 0.5$	IQL expectile	0.7	IQL temp.	3.0
L1	0.01	—				

Table 2: Implementation details for PPF experiments. Here d_n denotes the noise dimension.

5.2 Evaluation Metrics

We report four percentage-based metrics under nonlinear corruption and combine them into a composite score. Let $\mathcal{C}$ denote all evaluated noise configurations and $\mathcal{C}_{\text{hard}}$ the hardest subset.

- **Global Mean%:** average performance relative to the clean-state oracle, $\text{Global Mean\%} = \frac{1}{|\mathcal{C}|} \sum_{(d,\sigma) \in \mathcal{C}} \frac{\text{Score}(d,\sigma)}{\text{Oracle Mean}} \times 100$.
- **Hard Mean%:** performance on the most difficult configurations, $\text{Hard Mean\%} = \frac{1}{|\mathcal{C}_{\text{hard}}|} \sum_{(d,\sigma) \in \mathcal{C}_{\text{hard}}} \frac{\text{Score}(d,\sigma)}{\text{Oracle Mean}} \times 100$, where $\mathcal{C}_{\text{hard}} = \{(d,\sigma) \mid d \in \{0.75 d_{\text{true}}, d_{\text{true}}\}, \sigma \in \{1.5, 2.0\}\}$ and d_{true} is the true state dimension of the environment.
- **Retention:** how much performance is preserved as the noise scale increases, $\text{Retention} = 100 - \frac{\text{Score}(\sigma=0.5) - \text{Score}(\sigma=2.0)}{\text{Score}(\sigma=0.5)} \times 100$.
- **Win Rate:** the percentage of configurations where a method beats both the Plain Encoder and raw-noisy baselines, $\text{Win Rate} = \frac{|\{(d,\sigma) \in \mathcal{C} \mid \text{Score} > \text{Plain} \wedge \text{Score} > \text{RawNoisy}\}|}{|\mathcal{C}|} \times 100$.

The final composite score is

$$\text{Composite} = 0.30 \cdot \text{Global Mean\%} + 0.35 \cdot \text{Hard Mean\%} + 0.20 \cdot \text{Retention} + 0.15 \cdot \text{Win Rate}.$$

We assign the largest weight to Hard Mean% because severe corruption is the clearest robustness test.

5.3 Results

5.3.1 Main Results

Stage 1 We first evaluate all six disentanglement regularizers on HalfCheetah-medium-v2. Stage 1 is a broad nonlinear stress test. Figure 2 shows that all methods degrade as corruption strengthens, with disentanglement helping mainly in the moderate-noise region. This motivates the more controlled Stage 2 comparison.

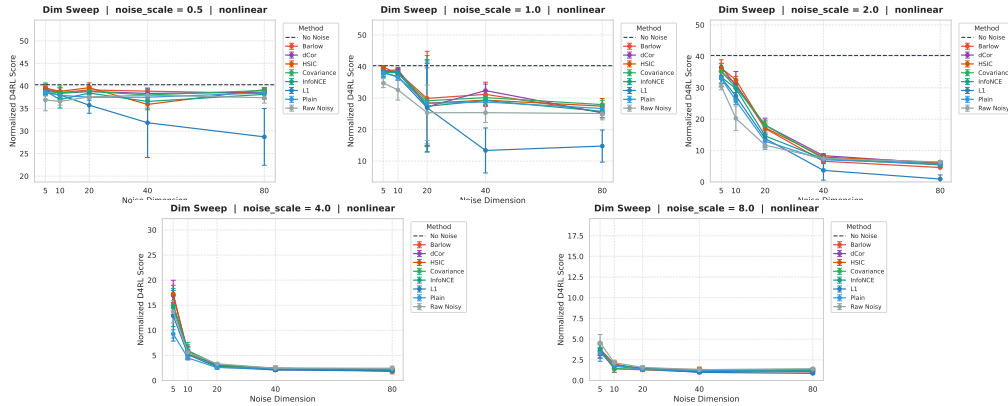


Figure 2: Stage 1 stress-test results on HalfCheetah-medium-v2 under nonlinear corruption.

Stage 2 Stage 2 scales the noise dimension with the original observation dimension. Figure 3 shows that performance decreases as either the noise scale or the noise dimension increases, and several disentanglement methods remain above the Plain Encoder and raw-noisy baselines under moderate to strong corruption.

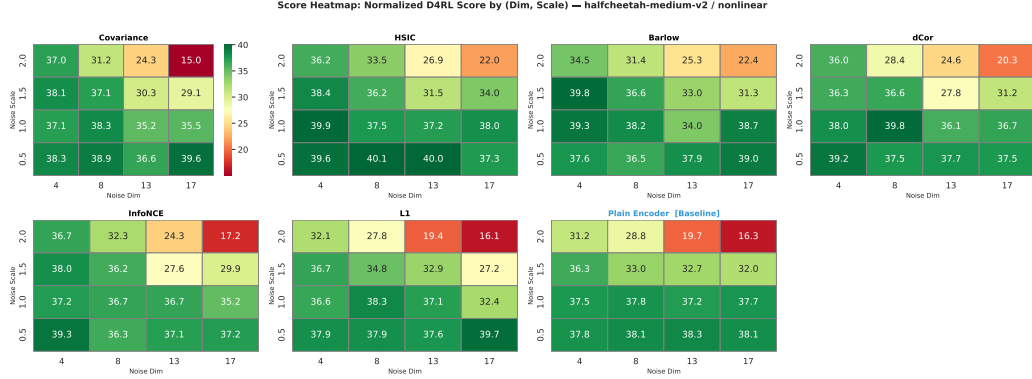


Figure 3: Score heatmap on HalfCheetah-medium-v2 under Stage 2 nonlinear corruption.

Figure 4 shows that HSIC ranks first, followed by Barlow Twins and distance correlation. We therefore select these three methods for Hopper and Ant.

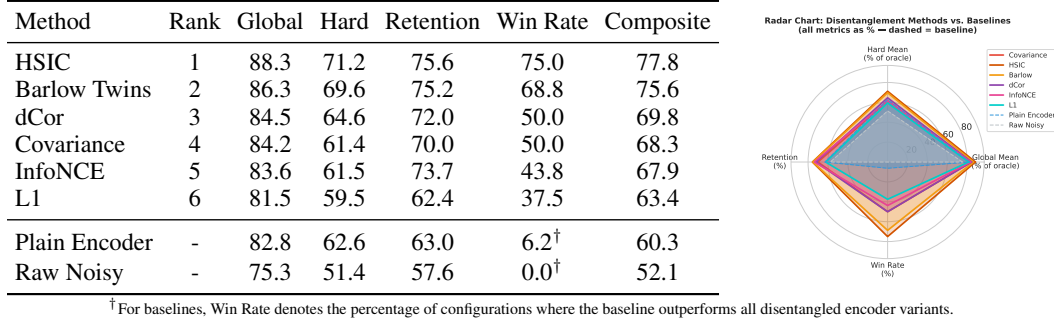


Figure 4: HalfCheetah-medium-v2 under Stage 2 nonlinear corruption.

These results suggest that regularizer effectiveness depends on the dependence structure induced by corruption. Under nonlinear corruption, $\text{Cov}(s, n) = 0 \not\Rightarrow s \perp n$, so even when clean state s and nuisance noise n are linearly uncorrelated, nonlinear mixing can leave residual dependence in the encoded representation h . Methods that penalize only linear correlation may therefore miss this residual dependence. HSIC and distance correlation, which capture nonlinear dependence among the dimensions of h , perform well in this setting, while Barlow Twins also remains effective through cross-correlation reduction. By contrast, covariance whitening, L1, and InfoNCE still improve over the baselines but offer smaller and less consistent gains.

Stage 3 Stage 3 tests whether disentanglement still helps under simpler concat and project corruption. As shown in Figure 5, the performance gaps are much smaller than under nonlinear corruption: the raw-noisy baseline and the Plain Encoder already stay close to the no-noise reference, since task-relevant and nuisance factors are either explicitly separated or only linearly mixed. Disentanglement therefore provides limited gains and can even hurt through over-regularization, confirming that it is most useful when corruption creates nonlinear dependence between task-relevant and nuisance information.

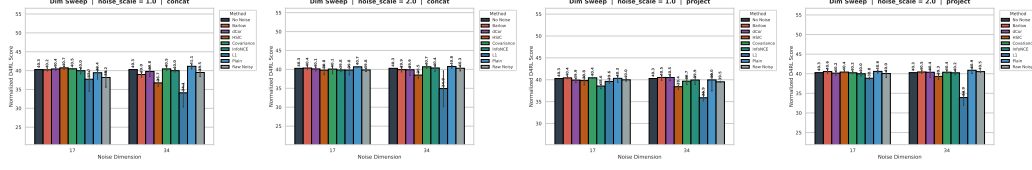


Figure 5: Stage 3 concat-corruption and project-corruption results on HalfCheetah-medium-v2.

5.3.2 Cross-Environment Results

Hopper-medium-v2. Figure 6 shows the score heatmap.

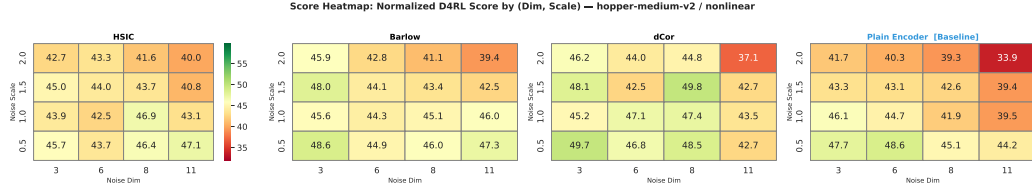
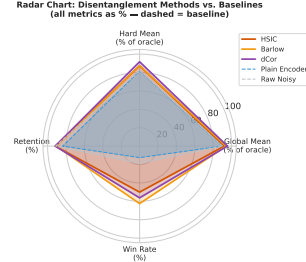


Figure 6: Score heatmap on Hopper-medium-v2 under nonlinear corruption.

Figure 7 confirms that distance correlation performs best overall, with the top Global Mean, Hard Mean, Retention, and Composite Score. Barlow Twins and HSIC remain competitive, but are less consistent on Hopper. Notably, the Plain Encoder (74.0) already underperforms Raw Noisy (76.8), so on Hopper the gain comes specifically from the disentanglement regularizer rather than from privileged supervision alone.

Method	Rank	Global	Hard	Retention	Win Rate	Composite
dCor	1	95.5	91.7	91.7	56.2	87.5
Barlow Twins	2	94.0	87.5	90.6	62.5	86.3
HSIC	3	92.1	87.3	91.6	50.0	84.0
Plain Encoder	-	89.6	81.6	83.5	12.5 [†]	74.0
Raw Noisy	-	89.9	82.1	91.2	18.8 [†]	76.8



[†]For baselines, Win Rate denotes the percentage of configurations where the baseline outperforms all disentangled encoder variants.

Figure 7: Hopper-medium-v2 under nonlinear corruption.

Ant-medium-v2. Figure 8 shows that Ant is much harder than Hopper: all methods degrade sharply as the noise dimension and scale increase, and disentanglement gains are far less consistent.

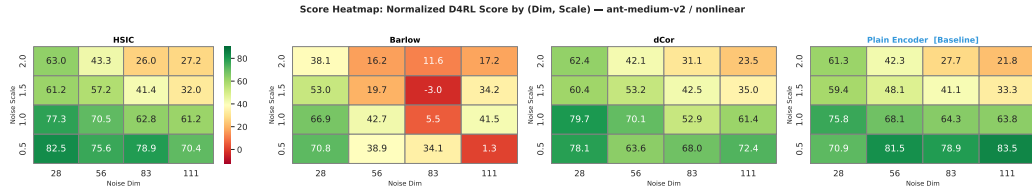


Figure 8: Score heatmap on Ant-medium-v2 under nonlinear corruption.

As shown in Figure 9, Raw Noisy achieves the highest composite score overall—a reversal absent on HalfCheetah and Hopper. Even the Plain Encoder (47.0) underperforms Raw Noisy (58.3), suggesting that the privileged pretraining itself struggles on Ant before any disentanglement is applied. This may

be because Hopper and HalfCheetah have similar original state dimensions, 11 and 17 respectively, so HalfCheetah-tuned hyperparameters can transfer more reasonably to Hopper. In contrast, Ant has a much higher original dimension of 111. Since our noise dimensions are set proportionally to the original state dimension, the total input dimensionality also becomes much larger on Ant, making the same regularization settings poorly calibrated. For Barlow Twins, the penalty scales roughly as $O(d_{\text{latent}}^2)$, so its independence weight should be much smaller on Ant. For HSIC, the fixed RBF bandwidth $\sigma = 1.0$ may be too small in the high-dimensional latent space. For dCor, although $\mathcal{L}_{\text{dCor}} = \text{dCor}(h)$ is normalized, the pairwise-distance structure can still change substantially in Ant’s high-dimensional and strongly coupled dynamics, so λ_d may also require retuning. These results suggest that cross-environment transfer requires re-calibrating both the encoder capacity and method-specific regularization parameters according to environment dimensionality and state structure.

Method	Rank	Global	Hard	Retention	Win Rate	Composite
HSIC	1	69.3	37.7	51.9	37.5	50.0
dCor	2	66.8	39.4	56.4	18.8	47.9
Barlow Twins	3	36.4	17.9	57.3	0.0	28.6
Plain Encoder	-	68.7	36.9	48.6	25.0 [†]	47.0
Raw Noisy	-	72.9	48.2	55.3	56.2 [†]	58.3

[†] For baselines, Win Rate denotes the percentage of configurations where the baseline outperforms all disentangled encoder variants.

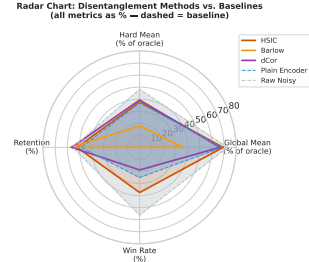


Figure 9: Ant-medium-v2 under nonlinear corruption.

Due to time constraints, we only conduct a preliminary sensitivity analysis for Barlow Twins. Figure 10 shows that the original HalfCheetah-tuned weight $\lambda_d = 0.05$ gives a near-zero score of -1.7 , while a smaller weight $\lambda_d = 0.005$ improves the score to 75.6, above both the Plain Encoder (69.4) and Raw Noisy (72.0) baselines. This suggests that Barlow Twins is highly sensitive to λ_d on Ant, and that its failure is more likely due to miscalibration than to the regularizer itself.

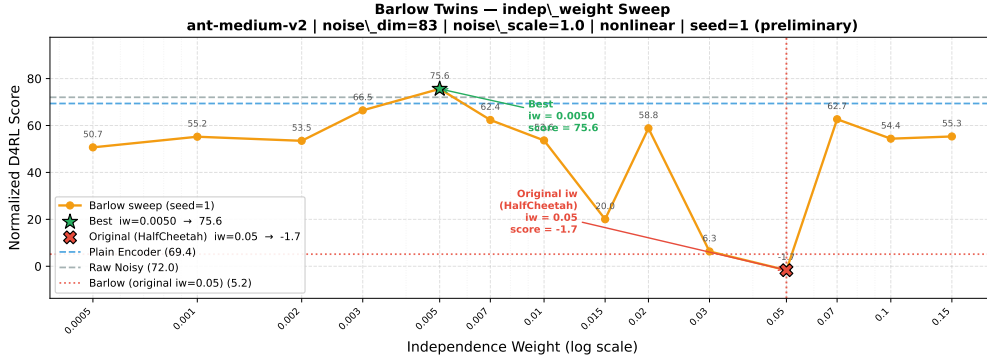
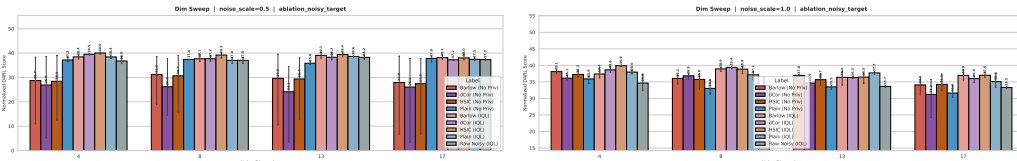


Figure 10: Preliminary Barlow Twins independence-weight sweep on Ant-medium-v2.

5.3.3 Ablation Results

No Privileged Supervision Ablation. This ablation replaces clean-state reconstruction with noisy-observation reconstruction while keeping other settings fixed.



5.3.4 External Method Comparison

We compare PPF-based methods with external baselines under the same HalfCheetah-medium-v2 nonlinear-corruption setting, fixing the noise scale and sweeping over the noise dimension.

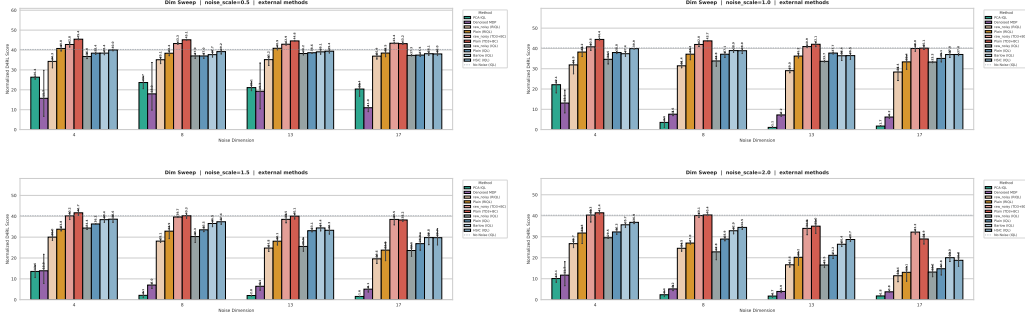


Figure 14: External method comparison on HalfCheetah-medium-v2 under nonlinear corruption.

Figure 14 shows that simple external strategies are insufficient under nonlinear corruption. PCA-IQL and Denoised MDP perform poorly in most settings, indicating that linear preprocessing or self-supervised representation learning without privileged clean-state supervision cannot reliably remove nonlinear nuisance factors. RIQL is more stable but does not consistently outperform PPF-based IQL methods. TD3+BC-based variants achieve strong results, especially under mild to moderate corruption, suggesting that the downstream offline RL backbone also plays an important role. Overall, PPF-style representation learning remains competitive with external robustness strategies, but its benefit is complementary to the choice of downstream RL algorithm.

6 Conclusions

We proposed PPF, a Privileged Pretraining Framework that uses clean-state supervision only during encoder pretraining and freezes the encoder for downstream offline RL on corrupted observations. Our results show that disentanglement is most useful under hard nonlinear corruption, while simpler concat and project corruption often need little extra regularization. HSIC, dCor, and Barlow Twins are generally the strongest methods on HalfCheetah and Hopper, and on Hopper their gain comes mainly from the disentanglement regularizer rather than from privileged supervision alone, but they require environment-specific tuning to work well on Ant. The ablations show that privileged clean-state or dynamics-aware supervision is important for anchoring the latent space; without it, disentanglement can become unstable or even hurt performance. External comparisons further suggest that PPF-style representation learning is complementary to the downstream offline RL backbone. Overall, disentangled representation learning can improve robustness in offline RL, but its benefit depends on corruption structure, supervision quality, environment dimensionality, and tuning.

7 Limitations and Future Work

- Some experiments use only three seeds, and the Ant Barlow sweep is preliminary with a single seed; future work should increase seeds and repeat sensitivity analysis across all methods.
- Hyperparameters tuned on HalfCheetah do not always transfer to high-dimensional environments such as Ant; future work should tune λ_d and kernel bandwidths per environment.
- The main PPF experiments still rely primarily on IQL and the backbone ablation only covers TD3+BC; future work should also evaluate PPF with RIQL and other downstream offline RL backbones to better separate representation-level and algorithm-level robustness.
- Concat and project corruption are relatively simple; future work should extend them to more environments and stronger settings.
- The current corruption model is synthetic; future work should study more realistic sensor-noise models and real-world perturbations.

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